KUSH PATEL

Data Engineer | ELT Pipelines · Lakehouse · Data Quality · Migration | SQL · PySpark · Microsoft Fabric · Azure Data Factory · Databricks

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Open to remote and relocation across Canada · Legally eligible to work in Canada · Available immediately

SUMMARY

- Data engineer with 4+ years building and running pipelines on Azure: as sole analyst at a perishable-goods distributor, brought ERP, Square POS, workforce and REST API feeds into Bronze/Silver/Gold layers in Microsoft Fabric and Azure Data Factory/Synapse, in SQL and PySpark.
- Treats data quality as part of the pipeline, not a report on it: schema contracts at ingestion, quarantine and replay for bad rows, gates that stop a load, and reconciliation at the feed that cut recurring discrepancies an estimated ~45%.
- Migrated a legacy SQL Server, SSIS and SSRS estate to Fabric without breaking a report, validating cutover with row counts, checksums and control totals; in public, the same discipline as a row-for-row GO/NO-GO gate across 107 artifacts.

TECHNICAL SKILLS

Pipelines & Orchestration: ETL and ELT in Azure Data Factory, Microsoft Fabric Data Pipelines, notebooks and Dataflows Gen2, Azure Synapse, SSIS, SQL Server Agent, GitHub Actions schedules, Databricks jobs; Kafka (Redpanda) streaming with event-time windows; Airflow DAG checked in CI, not run in production

Processing & Storage: SQL (T-SQL, window functions, CTEs), PySpark, Python (pandas, requests), Delta Lake, Microsoft Fabric (Lakehouse, Warehouse, OneLake), Databricks (Unity Catalog, SQL warehouse), Snowflake (stages, COPY INTO, dbt Projects on Snowflake), SQL Server, PostgreSQL, DuckDB, Parquet

Ingestion: REST APIs (including the Synerion time-and-attendance API), ERP and POS extracts, public data APIs and file feeds, incremental and CDC-style loads, schema-drift handling

Data Quality & Reliability: schema contracts, row quarantine and replay, quality gates, reconciliation and control totals, referential-integrity and freshness tests, injected-failure tests, pipeline alerting and exception logs, idempotent reruns

Modelling & Serving: Kimball star schemas, conformed dimensions, SCD Type 1/2, dbt Core (contracts, incremental, snapshots), Power BI semantic models with RLS/OLS

Delivery: Git, GitHub Actions CI, Docker, Terraform, Kubernetes (kind, in CI), pinned dependencies with an SBOM, pytest, cutover runbooks with disable-not-delete rollback

PROFESSIONAL EXPERIENCE

Two Rivers Specialty Meats

Dec 2023 – Jun 2026 · North Vancouver, BC

Data Analyst — Operations, Logistics & Enterprise Reporting

- Engineered cloud ELT in Microsoft Fabric and Azure Data Factory/Synapse using SQL and PySpark, with incremental loads, parameterised transforms and structured error handling.
- Integrated ERP, Square POS, workforce (Synerion API) and REST feeds into curated Bronze/Silver/Gold datasets serving analysis, BI dashboards and operational reporting, as the company's sole analyst.
- Owned the legacy SQL Server estate through migration: maintained SSIS packages and SSRS paginated reports while refactoring priority pipelines to Fabric notebooks, validating cutover with row counts, checksums and control totals so no reporting was disrupted.
- Built automated data-quality tests across the mart (schema, completeness, uniqueness, referential integrity) with anomaly thresholds, exception logging and pipeline alerting.
- Built reconciliation controls comparing source against curated totals for GL/P&L and inventory valuation, cutting discrepancies an estimated ~45% and reporting errors ~30% by fixing issues at the feed (internal figures, directional rather than audited).
- Designed the enterprise data mart (star schemas, conformed dimensions, SCDs) and served it through a Power BI semantic model secured with RLS/RBAC and column masking.
- Delivered FEFO and lot-level traceability and SKU-level demand forecasting as Python web apps for procurement.

Shivam Investments

Dec 2020 – Jul 2022 · Vadodara, India (hybrid)

Financial Analyst — Data & Reporting

- Automated recurring data preparation with Python (pandas) and SQL views feeding Power BI, replacing manual spreadsheet consolidation and cutting manual effort an estimated ~40%.
- Reconciled GL/P&L figures across sources and maintained traceable reporting for portfolio and risk stakeholders.

SELECTED PROJECTS — public code and CI-tested results; synthetic data is labelled

Supply Chain Control Tower — PySpark, Delta Lake, Databricks, SQL, Power BI · synthetic data
github.com/KushPatel29/supply-chain-control-tower · live: kush-network-risk-decision-room.streamlit.app
A medallion pipeline defended in three layers: schema contracts before Bronze, row quarantine with replay, and a quality gate on Gold, each proven in CI by injecting the failure it exists to catch. Benchmarked at 10M rows and run as a Databricks job, with all 36 Gold totals matching the local build to the cent.

Legacy-to-Fabric Migration — SSIS, SSRS, PySpark, Microsoft Fabric, Python · synthetic data
github.com/KushPatel29/legacy-to-fabric-migration · live: kush-asset-management-decision-board.streamlit.app
Cutover that has to prove itself: the move from SSIS/SSRS to Fabric waits on a row-for-row GO/NO-GO gate across 107 artifacts comparing row counts, keys, values, control totals and checksums. The gate is cent-exact and conjunctive, an empty output never passes, and tests break the data on purpose to prove it catches dropped rows, shifted values and errors that cancel out.

Canada Wildfire Risk — Python, requests, Parquet, GitHub Actions · real public data
github.com/KushPatel29/wildfire-prediction-app · live: wildfire-prediction-app1.streamlit.app
A scheduled pipeline that keeps running: pulls CWFIS station observations and an Open-Meteo forecast twice a day since September, rebuilds a seven-day forecast for 753 grid cells, and publishes it to a release and a public data branch. A weekly job grades every published forecast against satellite hotspots and commits the result. Found and fixed an index-alignment bug that had put 97.7% of fires in the wrong cell.

Real-time Transaction Monitoring — Kafka (Redpanda), Python, Terraform, Kubernetes · synthetic data
github.com/KushPatel29/aml-transaction-monitoring
A stream held to its batch: two windowed rules run in event time over 100,795 deliveries that arrive late and twice; at-least-once delivery with idempotent, window-keyed alerts on a compacted topic. Terraform deploys the broker, a two-replica consumer group and its jobs to a Kubernetes cluster in CI, a pod is killed mid-stream, and the run passes only if the stream flags exactly what the batch flags: 51 of 51 and 1,270 of 1,270.

Supply Chain Analytics — dbt — dbt Core, Databricks, Snowflake, DuckDB · synthetic data
github.com/KushPatel29/supply-chain-analytics-dbt
Same project, three engines: built on DuckDB, on a Databricks serverless SQL warehouse and inside Snowflake with dbt Projects on Snowflake: 180 of 180 nodes green and all 41 mart totals equal across all three, with contracts, incremental facts and SCD2 snapshots.

EDUCATION

Northeastern University — Master of Professional Studies in Analytics (Applied Machine Intelligence), GPA 3.76/4.0 · Vancouver, BC — **Sep 2022 – Jun 2024**

Gujarat Technological University — Bachelor of Engineering, Computer Science · India — **Aug 2017 – Aug 2021**

CERTIFICATIONS

Databricks Lakehouse Fundamentals · UC Davis SQL for Data Science · IBM Python for Data Science, AI & Development · Power BI Data Modelling with DAX · Google Data Analytics Professional Certificate. Preparing for the Microsoft DP-600 (Fabric Analytics Engineer) exam.